INORGANIC CUMULATIVE EXAM
APRIL 27, 1991

NOTE: If you were carrying out a departmental teaching assignment during the 2:30-3:30 PM period on April 25, 1991, you may take the alternate inorganic cume that follows this test.

1. During the 4/25/91 seminar by Professor Lipschutz (University of California, Santa Barbara), several terms were presented. Give concise statements (including chemical formulas where appropriate) to define or illustrate the following:

   a) lower order cuperate
   b) higher order cuperate
   c) transmetallation
   d) asymmetric coupling

2. What is the oxidation state of copper in these compounds?

3. Give an example of a reaction other than biphenyl coupling which can be carried out using cuperate reagents.

4. Distinguish between catalytic and stoichiometric organometallic reagents. To which category do cuperate reagents belong?

5. Describe the use of cuperate reagents in biphenyl coupling reactions. Discuss their use in syntheses of asymmetric coupling products. Present as much detail as possible about the method.

6. Describe the proposed mechanism that accounts for the variation of isomer yields of A-A, A-B, B-B (A,B = substituted phenyl) with temperature for biphenyl coupling.

7. Discuss a possible extension for the synthetic method discussed in #6 above.

8. Describe the new evidence for the nature of Gilman reagents presented at the seminar.

ALTERNATE INORGANIC CUMULATIVE EXAM
APRIL 27, 1991

Note the instructions below:

BORON CHEMISTRY - Answer any 7 of the following

1. Discuss a method for the preparation of elemental boron of purity > 95%.

2. Give an explanation for the order $BF_3 < BCl_3$ for Lewis Acid strength (actually this is the correct order!).

3. Make up any reasonable cyclic borate ion: sketch it to show all bonds and give the total negative charge.

4. Give the formulas for five neutral boron hydrides.

5. Sketch the expected ^{11}B nmr spectrum for B_5H_9.

6. Label the following as closo, nido or arachno, and name or describe the polyhedron on which the structure is based:

   $B_5H_7^{-2}$, $B_8H_{10}^{-4}$, $B_9H_9^{-2}$

7. Label the following as closo, nido or arachno, and name or describe the polyhedron on which the structure is based:

   $B_{10}C_2H_{12}$, $B_9C_2H_{12}^{-}$, $B_{10}SH_{10}^{-2}$

8. Give a reasonable preparation for borazine ("B-N benzene"). If you prefer, give the preparation for a trimethyl trichloro borazine.

9. Discuss reasons why the borazine ring is chemically more reactive than benzene.

10. Sketch the structure of diborane and describe its bonding.

NAME*:_____

I was teaching in _____ _____ during the April 25
(course) (section)

departmental seminar.

* This form will be removed before the exam is graded

(Detach at the double line and place in exam book)